

DESIGN AND PROTOTYPING OF POST-QUANTUM AUTHENTICATION FOR SOME/IP SERVICE DISCOVERY

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Abstract

This research addresses the lack of quantum resistant security in the Service Discovery (SD) phase of the Scalable service-Oriented Middleware over IP (SOME/IP) protocol. The current automotive architectures utilize unauthenticated multicast messages for service provisioning, which makes it vulnerable to spoofing and Man-in-the-Middle attacks. Post-Quantum Cryptography (PQC) signatures exceed the Maximum Transmission Unit (MTU) of automotive Ethernet, which prevents direct integration into protocols. This project proposes a new, stateless protocol extension to implement PQC without violating strict boot-time latency constraints. This report discusses the state of the art, the "MTU collision" problem, and proposed architecture.

Key words: Automotive Security, SOME/IP, Post-Quantum Cryptography, Service Discovery, Ethernet, Stateless Authentication.

1. Introduction

The automotive industry is moving away from old hardware-based connections (like CAN bus) toward software-based networks; this change is necessary because modern features like self-driving cars and advanced driver assistance systems (ADAS) need to send a lot of data very quickly. To handle this, the industry uses the SOME/IP (Scalable service-Oriented Middleware over IP) to communicate over Ethernet.

Vehicle's computers (ECUs) don't just send raw signals, they offer specific "services." For example, a camera ECU offers a "Rear Camera Service" that other parts of the car can find and use. To make this work, they use a protocol called SOME/IP Service Discovery (SOME/IP-SD) [1]. The SOME/IP protocol was designed for trusted, closed networks. It broadcasts critical messages, like "Offer Service" or "Find Service," to the whole network without authentication of the sender.

The lack of security in SOME/IP makes the system vulnerable towards Man-In-The-Middle and spoofing attacks. An attacker can connect to a car's diagnostic port, or hack into the infotainment system, then send fake "offer service" messages, and because the system is designed to accept the reply it gets, it accepts the service offer from the hacker. The vehicle's safety systems can be connected to a hacker instead of real ECUs, which presents critical dangers to the vehicle and the passengers such as disabling the brakes, or feeding false data to autopilot systems.

Standard security methods, such as RSA or ECDHE, are not suitable for fixing this issue for two reasons: they are too slow, which does not work in a modern car that needs to boot up and be ready in less than 50 milliseconds; and secondly, cars' lifetime on the road lasts for 15 years or more. Using current methods makes the protocol in danger of the "store now, decrypt later" threat; attackers could record traffic today and break it in the future with quantum computers. We need a solution that protects the cars against future threats right now.

1.1. Research Scope and Objectives

This research's focus is designing a "Post-Quantum Authentication Extension" for the SOME/IP-SD protocol. The problem I am addressing is "MTU Collision"; the quantum resistant signatures defined by NIST are too big (2KB-50KB), while the standard automotive ethernet packets are small. The signatures do not fit inside the packets.

In order to solve this, I propose the following:

Architecture: Design a "fragmentation" method to split the large signatures into smaller pieces so they can be sent over the network without breaking the protocol.

Algorithm Selection: Choose a cryptographic algorithm that is small enough to minimize the traffic, but fast enough for quick authentication

Prototyping: Build a simulation using the open-source library "someipy" to test if the design is both secure and fast enough for real cars.

2. Theoretical Background

To understand the solution, we first need to understand how the current protocol works and why timing is such a strict constraint.

2.1. Protocol Mechanics and Timing

SOME/IP-SD operates on a publish/subscribe basis using Multicast Groups rather than simple broadcasts; it runs over UDP on port 30509. A service, for example rear sensors or cameras, joins a specific multicast group to 'OfferService', this allows other ECUs to subscribe to that specific stream without clogging the rest of the network.

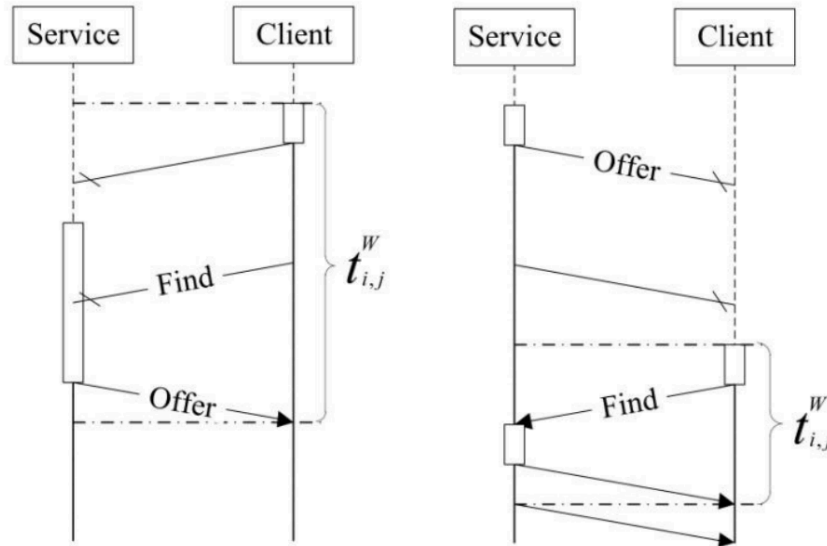


Figure 1: SOME/IP-SD Message Exchange Sequence and Initial Wait Phase [2]

Speed is a critical factor for this connection. When a car starts up, there is a "burst" of messages in the first 100-200 milliseconds. This is known as the "Initial Wait" and "Repetition" phases [2]. Therefore, it was concluded in the research by Seyler et al. that there can be no compromise in the speed of this process. Any security check added, such as verifying a signature, must happen almost instantly, often less than 10 milliseconds. If not, the vehicle's safety systems will be delayed, which is unacceptable [2].

3. State of the Art

The research on SOME/IP security has gone through three main stages: identifying the flaws, trying to fix them with standard cryptography, and now dealing with the new threat of quantum computing.

3.1. Verification of Vulnerabilities

SOME/IP is currently proven to be insecure. It was proved mathematically by Zelle et al. in their research where they identified a specific flaw called a "race condition." Because a vehicle's computer accepts the first message it receives, an attacker only needs to be slightly faster than the real server in order to trick the system [3]. A "Man-in-the-Middle" attack works even if the network cables themselves are secured (using MACsec) because the vulnerability is in the software logic, not the physical wires [3].

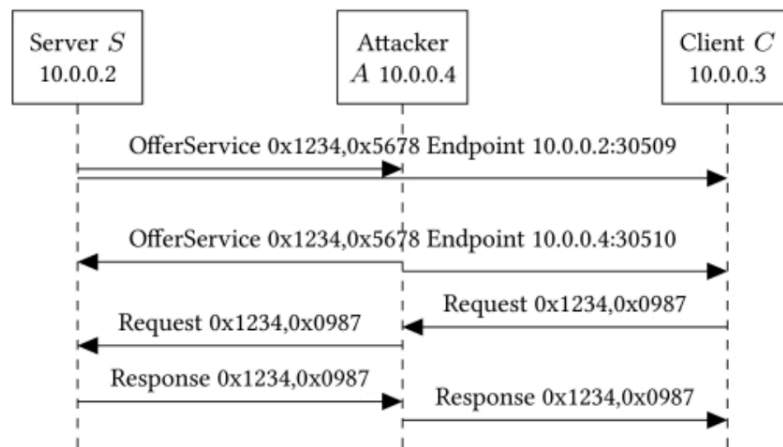


Figure 2: Sequence Diagram of a Man-in-the-Middle (MITM) Attack on SOME/IP [3]

A "race condition" is not the only vulnerability SOME/IP faces. An attacker can send a single fake "Stop Service" message to disconnect safety features instantly [4]. Moreover, if the network is flooded with junk requests, an attacker can cause the network to run out of memory and crash in a Denial of Service attack [5].

3.2. Limitations of Detection and Classical Prevention

In order to prevent the attacks mentioned, Artificial intelligence methods were implemented; deep learning was used to spot fake messages with high accuracy. However, it only spots the spoofed message after it had been processed [6], which is not sufficient or suitable for a safety-critical system.

Standard encryption methods, such as TLS, were proven to be secure; however, it is too slow. Setting up the secure session for every service creates a "boot storm" of data that delays the vehicle's startup [7]. Another method was using a central DNS server to manage security, which solves the speed issue, but introduces another problem, which is having a single point of failure for the system: if the server crashes, the entire system crashes.

3.3. The Post-Quantum "MTU Collision"

The biggest issue in securing SOME/IP-SD and the problem we are addressing is the "MTU Collision".

Rathweg et al. tested the new NIST-standardized Post-Quantum algorithms on automotive chips and found that there is a physical mismatch between them[9].

Constraint: A standard Ethernet packet can only hold about 1500 bytes.

Conflict: The new secure signatures (Dilithium) are very large, between 2400 and 4600 bytes[9].

The Result: A signature this big cannot fit inside the packet.

Other algorithms don't work well either for various reasons, SPHINCS+ takes seconds to verify, making it too slow, and FALCON is fast but requires complex hardware that many vehicle computers don't have[9].

3.4. The Fragmentation Gap

In many Post-Quantum Computing implementations in modern protocols, splitting the message or fragmentation is the approach towards making traditional protocols Post-Quantum resistant, however, if we rely on standard network splitting (IP fragmentation), it becomes unreliable due to the fact that if one piece of the message is lost, the entire process fails[10], which is why a more suitable and efficient way is required to handle the large PQC signatures.

4. Proposed Architecture and Feasibility

The literature review confirms a gap: nobody has designed a specific way to send large quantum signatures over SOME/IP reliably. I am proposing a "Stateless Application-Layer Extension" to fix this.

4.1. Concept

I propose a method of fragmenting at the application layer that uses timing as a feature rather than a constraint, and instead of inventing a new protocol, I will utilize the Type-Length-Value (TLV) structures within the existing 'Options' field of the SOME/IP header to carry the security data

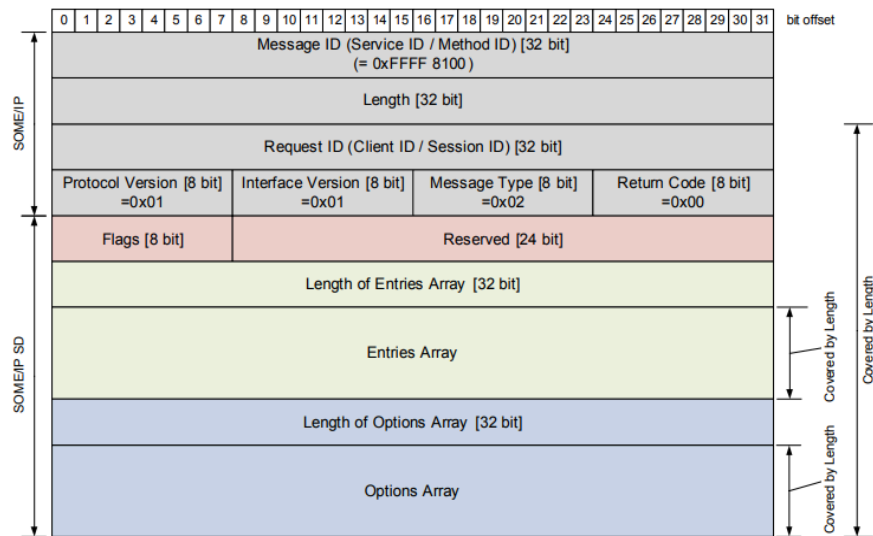


Figure 3: Standard SOME/IP-SD Header Format showing the "Options Array" field where the security payload will be inserted [1]

Fragmentation: Split the large 2.4KB Dilithium signature into two smaller 1.2KB fragments.

Transport: Attach the first fragments to the first "Offer" message, then I will attach the second fragment to the repetition message that automatically follows 10ms later.

Reassembly: The receiver waits until it has both fragments, reconstructs the signature, and verifies it, this adds security without creating extra network traffic.

6. Second Semester Plan

In the next semester, I plan to move from theoretical research to the actual design of the solution, the main goal is to define exactly how the SOME/IP packet header needs to change to carry the new security data, I will also create a basic simulation using Python and “someipy” to test this design, this will help me prove that large signatures can be split into smaller pieces (fragmentation) and successfully put back together by the receiver without losing data.

7. Conclusions

This report confirms that securing automotive networks against quantum threats is currently facing a number of challenges; classical security methods [7, 8] are not strong enough for the future, but the new Post-Quantum algorithms [9] are physically too large for standard Ethernet packets; moreover, relying on standard network fragmentation [10] causes reliability issues. Therefore, the only viable solution is to design a custom fragmentation strategy specifically for the SOME/IP application layer. My future work will focus on implementing this design and benchmarking its performance to prove it works.

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